Irrational equations

Squaring is the only way forward and the only source of false roots — so the check is not optional, it is the method.

LESSON 40–42 | 3 × 40 MIN | ALGEBRA 10, §2.5 | P1 · 1.5 (EXTENSION)

LESSON CLOCK

14' | 24' | 30' | 24' | 36' | 7'

Two conditions, not one	14 min
One radical	24 min
Two radicals	30 min
Substitutions	24 min
Practice	36 min
Homework	7 min

LEARNING OBJECTIVES

- State the two conditions an irrational equation imposes before squaring.
- Solve an equation with one radical by isolating and squaring.
- Solve an equation with two radicals by squaring twice.
- Check every candidate root in the original equation.

TEXTBOOK

National	pp. 147–162
Cambridge	Pure Mathematics 1, pp. 12–17
Workbook	P1 Exercise 1E

01 Explanation

Two conditions before you start

$\sqrt{A} = B$ REQUIRES BOTH

1 $A \geq 0$ — the radicand cannot be negative. **2** $B \geq 0$ — a square root is never negative, so the other side cannot be either. Condition 2 is the one that is forgotten, and it is the one that creates false roots.

$\sqrt{x+3} = -2$ has no solution at all — the left side is never negative — but squaring gives $x + 3 = 4$ and the plausible-looking $x = 1$. Squaring destroys sign information, permanently.

$$\sqrt{a} \longrightarrow a^{1/2}$$

$$\sqrt[3]{a} \longrightarrow a^{1/3}$$

$$\sqrt[4]{a} \longrightarrow a^{1/4}$$

$$\sqrt[n]{a^m} \longrightarrow a^{m/n}$$

a root is a power — the same object, two notations

The graph of a square root lives entirely above the axis. Nothing below it can be an answer.

One radical

1. **Isolate** the radical on one side, alone.
2. Square both sides.
3. Solve the polynomial equation.
4. **Check every root** in the *original* equation.

$$\sqrt{2x+3} = x \Rightarrow 2x + 3 = x^2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow x = 3, -1$$

Check: $x = 3$ gives $\sqrt{9} = 3$ ✓. $x = -1$ gives $\sqrt{1} = -1$ ✗. Only $x = 3$.

ISOLATE FIRST, SQUARE SECOND

Squaring $\sqrt{x} + 1 = 4$ directly gives $x + 2\sqrt{x} + 1 = 16$ — a new radical, and more work. Move the 1 across first.

Two radicals

Arrange one radical on each side, square, then isolate whatever radical survives and square again:

$$\sqrt{x+5} - \sqrt{x} = 1 \Rightarrow \sqrt{x+5} = 1 + \sqrt{x}$$

$$x + 5 = 1 + 2\sqrt{x} + x \Rightarrow 4 = 2\sqrt{x} \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4$$

Check: $\sqrt{9} - \sqrt{4} = 3 - 2 = 1$ ✓.

WHY ONE ON EACH SIDE

$(a - b)^2$ still contains a cross term with both radicals; $(1 + \sqrt{x})^2$ contains only one. Arranging the equation before squaring halves the work.

Substitutions

When the same radical appears more than once, name it:

$$x + \sqrt{x} - 6 = 0 \text{ , } let\ t = \sqrt{x} \geq 0$$

Then $t^2 + t - 6 = 0$, so $t = 2$ or $t = -3$. The second is impossible because $t \geq 0$, so $\sqrt{x} = 2$ and $x = 4$.

The condition $t \geq 0$ is written down with the substitution, not remembered later. It is what removes the false root before any checking is needed.

02 Worked examples

EXAMPLE 1 **Solve $\sqrt{3x + 1} = x - 1$.**

1. Conditions: $3x + 1 \geq 0$ and $x - 1 \geq 0$, so $x \geq 1$.
2. $3x + 1 = x^2 - 2x + 1$
3. $x^2 - 5x = 0 \Rightarrow x = 0, 5$
4. $x = 0$ fails $x \geq 1$.

ANSWER

$x = 5$

EXAMPLE 2 **Solve $\sqrt{x + 7} - \sqrt{x} = 1$.**

1. $\sqrt{x + 7} = 1 + \sqrt{x}$

   One on each side.

2. $x + 7 = 1 + 2\sqrt{x} + x$

3. $6 = 2\sqrt{x} \Rightarrow \sqrt{x} = 3$

4. $x = 9$. Check: $4 - 3 = 1$ ✓

ANSWER

$x = 9$

EXAMPLE 3 **Solve $x - 5\sqrt{x} + 6 = 0$.**

1. Let $t = \sqrt{x} \geq 0$.

   $t^2 - 5t + 6 = 0$

2. $t = 2$ or $t = 3$, both allowed.

3. $\sqrt{x} = 2 \Rightarrow x = 4$

4. $\sqrt{x} = 3 \Rightarrow x = 9$

ANSWER

$x = 4$ and $x = 9$

03 Interactive model

Sketch both sides as graphs — the intersections are the true roots, and false roots are visibly absent.

Roots and powers

$$\sqrt[3]{64} = 64^{1/3} = \mathbf{4}$$

base a **64** | index n **3** | power m **1** | check: value^3 **64** | a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Irrational equation	Irratsional tenglama	Иррациональное уравнение
Radical	Radikal	Радикал
Radicand	Ildiz ostidagi ifoda	Подкоренное выражение
Isolate the radical	Radikalni ajratish	Уединить радикал
Squaring both sides	Kvadratga ko‘tarish	Возведение в квадрат
False (extraneous) root	Chet ildiz	Посторонний корень
Necessary condition	Zaruriy shart	Необходимое условие
Equivalent transformation	Teng kuchli almashtirish	Равносильное преобразование
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sqrt{A} = B$ requires:

- $A \geq 0$
- $B \geq 0$
- both
- neither

$\sqrt{x + 3} = -2$ has:

- $x = 1$
- $x = -7$
- no solution
- two solutions

The first step with $\sqrt{x} + 1 = 4$ is:

- square
- isolate the radical
- substitute
- cube

For $x + \sqrt{x} - 6 = 0$ set:

- $t = x$
- $t = \sqrt{x}$
- $t = x^2$
- nothing

After squaring you must:

- stop
- check each root in the original
- square again
- factorise

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sqrt{x} = 3$

ANSWER $x = 9$

2. Solve $\sqrt{x + 1} = 2$

ANSWER $x = 3$

3. Solve $\sqrt{2x} = 4$

ANSWER $x = 8$

4. Solve $\sqrt{x} = -1$

ANSWER no solution

5. Solve $\sqrt{x - 3} = 0$

ANSWER $x = 3$

6. Solve $\sqrt{5 - x} = 1$

ANSWER $x = 4$

7. Condition on x in $\sqrt{x - 2}$

ANSWER $x \geq 2$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{2x + 3} = x$

   ANSWER $x = 3$

2. Solve $\sqrt{3x + 1} = x - 1$

   ANSWER $x = 5$

3. Solve $\sqrt{x + 7} - \sqrt{x} = 1$

   ANSWER $x = 9$

4. Solve $x - 5\sqrt{x} + 6 = 0$

   ANSWER $x = 4, 9$

5. Solve $\sqrt{x + 4} = x - 2$

   ANSWER $x = 5$

6. Solve $\sqrt{x^2 - 3} = 1$

   ANSWER $x = \pm 2$

7. Solve $\sqrt{x} + 1 = 4$

   ANSWER $x = 9$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{x + 5} + \sqrt{x} = 5$

   ANSWER $x = 4$

2. Solve $\sqrt{2x + 5} - \sqrt{x - 1} = 2$

   ANSWER $x = 10$ or $x = 2$

3. Solve $x + \sqrt{x + 5} = 7$

   ANSWER $x = 4$

4. Solve $\sqrt{x^2 + 3x} = x + 1$

   ANSWER $x = 1$

5. Solve $2x - 7\sqrt{x} + 3 = 0$

   ANSWER $x = 9$ or $x = 0.25$

6. Solve $\sqrt{\sqrt{x}} = 2$

   ANSWER $x = 16$

7. Find all k for which $\sqrt{x} = x + k$ has exactly one solution

   ANSWER $k = 0.25$ or $k \le 0$

07 Homework

Homework — 6 tasks

Every answer must show the check written out in full.

1. Solve $\sqrt{4x + 1} = x - 1$.
2. Solve $\sqrt{x + 9} - \sqrt{x} = 1$.
3. Solve $x - 7\sqrt{x} + 12 = 0$.
4. Solve $\sqrt{2x - 1} = x - 2$.

5. Explain, in three sentences, why squaring can create a root the original equation never had.
6. Show that $\sqrt{x + 2} = -3$ has no solution without squaring.